Database Systems Assignment 3

Question 1:

1. List first and last names of doctors with more than 7 years experience:

$$\pi_{FirstName, LastName} (\sigma_{YearsOfExperience > 7} (Doctor))$$

2. Find patients who share the same primary doctor:

$$\pi_{P1.FirstName, P1.LastName} \left(\sigma_{P1.PrimaryDoctor_SSN = P2.PrimaryDoctor_SSN \land P1.SSN \neq P2.SSN} \left(\rho_{P1}(Patient) \times \rho_{P2}(Patient) \right) \right)$$

3. List cardio specialists:

$$\pi_{FirstName, LastName} (\sigma_{Specialty = 'Cardio'} (Doctor))$$

4. Find generic medicines under $50:

```
π_TradeName (σ_GenericFlag = True ∧ UnitPrice < 50 (Medicine))
```

5. Patients of Dr. John Smith:

```
π_Patient.FirstName, Patient.LastName (
  σ_Doctor.FirstName = 'John' ∧ Doctor.LastName = 'Smith'
  (Patient ⋈ Patient.PrimaryDoctor_SSN = Doctor.SSN Doctor)
)
```

6. Doctors with no patients:

```
π_FirstName, LastName (
  Doctor − π_Doctor.SSN, FirstName, LastName (
     Doctor ⋈ Doctor.SSN = Patient.PrimaryDoctor_SSN Patient
  )
)
```

7. Patients prescribed both Aspirin and Vitamin:

```
π_Prescription.Patient_SSN (
```

σ_PM1.TradeName = 'Aspirin' ∧ PM2.TradeName = 'Vitamin' ∧

PM1.PrescriptionId = PM2.PrescriptionId

((ρ_PM1(Prescription_Medicine) × ρ_PM2(Prescription_Medicine))

⋈ PM1.PrescriptionId = Prescription.Id

⋈ PM2.PrescriptionId = Prescription.Id

Prescription)

)

8. Patients prescribed Aspirin by Dr. John Smith:

π_Patient_SSN (

σ_TradeName = 'Aspirin' ∧ Doctor.FirstName = 'John' ∧ Doctor.LastName = 'Smith'

(((Prescription_Medicine ⋈ Prescription_Medicine.PrescriptionId = Prescription.Id Prescription)

⋈ Prescription.Doctor_SSN = Doctor.SSN Doctor))

)

9. Patients only treated by their primary doctor:

AllPatients = π_SSN, FirstName, LastName (Patient)

PatientsWithOtherDoctors =

  π_Patient.SSN, Patient.FirstName, Patient.LastName (

    σ_Patient.SSN = Prescription.Patient_SSN ∧ Prescription.Doctor_SSN ≠ Patient.PrimaryDoctor_SSN

    (Patient ⋈ Prescription)

)

π_FirstName, LastName (AllPatients - PatientsWithOtherDoctors)

10. Patients prescribed medicines under $100:

π_Patient.FirstName, Patient.LastName (

  σ_UnitPrice < 100 (

    (((Prescription_Medicine ⋈ Prescription_Medicine.PrescriptionId = Prescription.Id Prescription)

    ⋈ Prescription.Patient_SSN = Patient.SSN)

    ⋈ Prescription_Medicine.TradeName = Medicine.TradeName)

)

)

Question 2:

a. Employees in department 4:

π_Fname, Lname (σ_Dno = 4 (Employee))

b. Department 4 employees working on ProductY for >10 hours:

π_E.Fname, E.Lname (

  σ_E.Dno = 4 ∧ W.Hours > 10 ∧ P.Pname = 'ProductY'

  (((Employee E ⋈ E.Ssn = W.Essn Works_On W) ⋈ W.Pno = P.Pnumber Project P))

)

c. Employees supervised by Franklin Wong:

π_E.Fname, E.Lname (

  σ_S.Fname = 'Franklin' ∧ S.Lname = 'Wong'

  ((Employee E ⋈ E.Super_ssn = S.Ssn) ⋈ ρ_S(Employee))

)

d. Total hours per project:

```
π_P.Pname, SUM_Hours (
  γ_P.Pname; SUM(W.Hours)→SUM_Hours
  (Works_On W ⋈ W.Pno = P.Pnumber Project P)
)
```

e. Employees working on all projects:

```
π_E.Fname, E.Lname (
  Employee E –
  (π_E.Fname, E.Lname (
     (Employee E × Project P)
     – (Works_On W ⋈ E.Ssn = W.Essn ∧ P.Pnumber = W.Pno)
  ))
)
```

f. Employees not working on any project:

```
π_Fname, Lname (
```

Employee − π_E.Fname, E.Lname (Employee E ⋈ E.Ssn = W.Essn Works_On W)

)

g. Average salary per department:

π_D.Dname, AVG_Salary (

γ_D.Dname; AVG(E.Salary)→AVG_Salary

((Employee E ⋈ E.Dno = D.Dnumber Department D))

)

h. Average female salary:

π_AVG_Salary (

γ_AVG(E.Salary)→AVG_Salary (σ_Sex = 'F' (Employee))

)

i. Employees working on Houston projects but department not in Houston:

π_E.Fname, E.Address (

σ_P.Plocation = 'Houston' ∧ ¬DeptHasHouston

((Employee E ⋈ E.Ssn = W.Essn Works_On W ⋈ W.Pno = P.Pnumber Project P)

−

(Employee E ⋈ E.Dno = DL.Dnumber Dept_Locations DL ∧ DL.Dlocation = 'Houston')

)

)

j. Managers with no dependents:

π_Lname (

(Employee ⋈ Employee.Ssn = Department.Mgr_Ssn)

−

(Employee ⋈ Employee.Ssn = Department.Mgr_Ssn ⋈ Employee.Ssn = Dependent.Essn)

)

Question 3:

a. Ratings of sailors over 30:

$\pi_{rating} (\sigma_{age > 30} (Sailors))$

b. Days Dustin reserved boats:

$\pi_{day} (\sigma_{sname='Dustin'} (Sailors \bowtie Reserves))$

c. Colors of boats Dustin reserved:

$\pi_{color} (\sigma_{sname='Dustin'} (Sailors \bowtie Reserves \bowtie Boats))$

d. Sailors with rating ≥8 or reserved boat 103:

$\pi_{sid} (\sigma_{rating \geq 8} (Sailors)) \cup \pi_{sid} (\sigma_{bid=103} (Reserves))$

e. Sailors who never reserved red boats:

$\pi_{sname} (Sailors) - \pi_{sname} (\sigma_{color='red'} (Sailors \bowtie Reserves \bowtie Boats))$

f. Sailors who never reserved any boat:

$\pi_{sname} (Sailors) - \pi_{sname} (Sailors \bowtie Reserves))$

g. Sailors who reserved all boats:

π_sname (Sailors ⋈ (π_sid (Reserves) ÷ π_bid (Boats)))

h. Sailors who reserved all Interlake boats:

π_sname (Sailors ⋈ (π_sid (Reserves) ÷ π_bid (σ_bname='Interlake' (Boats))))

i. Sailors with rating higher than Bob:

π_sid (σ_rating > (π_rating (σ_sname='Bob' (Sailors))) (Sailors))

j. Highest-rated and oldest sailors:

-- Highest-rated sailors

π_sid (σ_rating = (π_MAX(rating) (Sailors)) (Sailors))

-- Oldest sailor

π_sname, age (σ_age = (π_MAX(age) (Sailors)) (Sailors))

Question 4:

Original table violates 1NF because:

- The Items column contains repeating groups

- Each cell should contain atomic values

- The table has non-atomic attributes

1NF requirements:

- All attributes contain atomic (indivisible) values

- No repeating groups of attributes

- Each entry contains a single value for each attribute

Decomposed Tables (1NF):

Orders (OrderID is PK)

- OrderID (PK)

- CustomerName

OrderItems (Composite PK: OrderID, ProductID)

- OrderID (FK to Orders)

- ProductID (FK to Products)

- Quantity

Products (ProductID is PK)

- ProductID (PK)

- ProductName

- SupplierName

- UnitPrice

Final 3NF Tables:

Orders

- Attributes: OrderID (PK), CustomerName

- Functional Dependency: OrderID → CustomerName

OrderItems

- Attributes: OrderID (FK), ProductID (FK), Quantity

- Functional Dependency: (OrderID, ProductID) → Quantity

Products

- Attributes: ProductID (PK), ProductName, SupplierName, UnitPrice

- Functional Dependencies:

  - ProductID → ProductName

  - ProductID → SupplierName

  - ProductID → UnitPrice